

## Indexer Cluster Lab

**IMPORTANT:** The **approximate build time for this lab is 15 minutes**. Please be patient. If the lab does not build after 30 minutes, try refreshing the page and rebuilding it again. If it does not build on the second attempt, contact [ellearn@splunk.com](mailto:ellearn@splunk.com).

### Lab Overview:

*Maximum Estimated Duration: 4 hours*

In this lab you will be connecting a "pre-existing" infrastructure formed of two un-clustered indexers, a search head and a monitoring console, then migrating this stand-alone structure to take on a new cluster master. This new cluster master will manage the indexers in a newly formed indexer cluster, connect the search head to the indexers and manage index replication. You will finish the lab by migrating the customer's data into the clustered bucket format so that the data can be replicated across the cluster.

Use <http://{{dnsname}}:8000> to access the WEB GUI of servers in this lab what come preinstalled with Splunk.

| Host Name        | Role                                             |
|------------------|--------------------------------------------------|
| Cftrl5-xxxx-sh1  | Search Head                                      |
| Cftrl5-xxxx-mc1  | Monitoring Console                               |
| Cftrl5-xxxx-idx1 | Indexer                                          |
| Cftrl5-xxxx-idx2 | Indexer                                          |
| Cftrl5-xxxx-cm1  | Cluster Master (no existing Splunk installation) |

### Configure the Management Console

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First you will set up your management console host (which also acts as the Deployment Server). This requires making the MC host a search head to all other instances in the lab environment.

#### Peer the management console to all available hosts

2. Navigate to Settings > Distributed Search > Search Peers > Add New.
3. Provide the IP address and management port, as well as the appropriate admin credentials to establish search peering.
4. Repeat for all other instances.

## Initialize the Cluster Master

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Now you will leverage both the base configs for clustering (URL below) and the deployment server to prepare the cluster master.

To download the base configs, click:

<https://drive.google.com/drive/folders/10aVQXjbgQC99b9InTvncrLFWUrXci3gz>

Install Splunk Enterprise and deploy appropriate base configs:

1. Install Splunk on the cluster master instance with the proper runtime user. Use the password provided in your Server list)
2. Using the base configs for indexer clustering, configure and install the appropriate apps in `$SPLUNK_HOME/etc/apps` on the cluster master.

Pay special attention to `replication_factor`; you will only have two indexer nodes available.

3. Restart the instance for these changes to take effect.
4. Observe that the CM's GUI reports that it is Cluster Master mode. At this time, it should report that it has no peers configured.
5. Add this new node as a distributed search peer to the monitoring console.

Before configuring peers, we'll ensure that the appropriate shared configuration is coming over from the Deployment Server directly to the Cluster Master.

6. Log in to the DS instance GUI, and go to Settings -> Forwarder Management.
7. For the cluster master host, create a serverclass and add the same apps that the indexer class has.

The cluster master acts as a proxy for sending indexer configuration settings to the indexers themselves.

8. Edit the `serverclass.conf` file directly to add `'stateOnClient = noop'` to each of the app entries for the indexer server class.

An “app” entry looks like `[serverClass:<classname>:app:<app>]`.

There is not a way to set this option directly via the Forwarder Management GUI.

9. Ensure that the cluster master has been correctly whitelisted in the cluster master serverclass, and appropriately blacklisted from the `all_hosts` serverclass.

10. After your command-line edits of `serverclass.conf`, you will have to manually issue the command to reload classes:  
`splunk reload deploy-server`
11. Return to the CLI of the cluster master, and `cd` into the `$SPLUNK_HOME/etc/master-apps` directory.
12. Observe the contents of that directory (you may have to wait a few minutes), and ensure that it matches the set of apps that was shipped to your (standalone) indexers. Make sure, however, not to include your `deploymentclient` app!

## Troubleshooting Suggestions

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If your configuration is not returning the expected results, troubleshoot by isolating the issue.

Verify the mode of the CM instance in the `[clustering]` section of `server.conf`  
`splunk btool server list clustering -debug`

Check on the Cluster Master to validate that it is acting as a deployment client; look for `$SPLUNK_HOME/var/run/serverclass.xml`. This file is a catalog of the server classes matched by the host, and the apps that were installed as a result.

## Migrate the Stand-alone Indexers

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1. Remove the indexers from the Deployment Server and join them to Cluster Master
2. Now convert your standalone indexers so that they will participate in an indexing cluster.

This real-world scenario will mimic what you will have to do for customers who are implementing indexer clustering to attain fault tolerance and high availability.

3. Begin by decoupling the indexer from the deployment server.

The easiest way to achieve this is to stop Splunk, remove the `deploymentclient` app, and restart Splunk.

It is important that the cluster master is the only source of configuration data for a clustered indexer. Splunk will disable this functionality when it starts up, and does not find a `deploymentclient.conf`.

**NOTE:** Removing apps that are managed by the deployment client / server interaction while Splunk is running may lead to confusing results. The client may refresh the app contents from its

local cache before you have a chance to restart Splunk. In particular, the `org_all_deploymentclient` app may be restored from cache, meaning that a `deploymentclient.conf` will be present when Splunk starts, enabling the client functionality once more. Therefore, it is important to stop Splunk before removing any apps managed by the deployment server.

4. Use the indexer-specific base cluster configs (note: multisite clustering is not in play at this time), placed in the `$SPLUNK_HOME/etc/apps` directory to prepare this host to be a clustered indexer.
5. Validate the `master_uri` setting, as well as the `pass4SymmKey`; this must match the one set for the master.
6. Use 9887 for your indexer replication port
7. Restart the indexer and wait for it to join the cluster
8. Repeat for the second indexer.

## Troubleshooting Suggestions

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If the indexer fails to join the cluster, it could be because the Indexer doesn't like the bundle provided by the master. However, there are other reasons why the member may not join, so consider the following steps below.

1. Verify that the master URI is correct. In the lab environment, it's likely to be a bare IP, and not the "cfrtl5-\*" type name that appears in the server list. Also note that it should not have a trailing slash.
2. Make sure that the bundle from the cluster master provides a consistent picture of the index configuration; this is critical for the peers to ensure that they can smoothly replicate data to the other hosts in the cluster. In 6.2 releases, the `_introspection` index is not included in the `_cluster` app's `indexes.conf` to include a "repFactor" setting.
3. Check the contents of the cluster bundle. Navigate to the cluster master's directory `$SPLUNK_HOME/var/run/splunk/cluster/remote-bundle`. Contained here are directories and files; `.bundle` files are tar archives. The filename ending in `.bundle_active` shows which bundle file / directory are current. The ten-digit number at the end of the bundle's name is the epoch time stamp when it was created. Look inside the directory indicated by the `.bundle_active` filename, and validate the contents. You can tweak `btool` to show the contents here; you must first symlink `$SPLUNK_HOME/etc/system` into the bundle's directory.
 

```
cd $SPLUNK_HOME/var/run/splunk/cluster/remote-bundle/<active bundle dir>
ln -s $SPLUNK_HOME/etc/system
$SPLUNK_HOME/bin/splunk cmd btool --dir=`pwd` <conf> <cmd>
```

4. Validate the pass4SymmKey matches host to host. This setting acts like a "secret handshake" between peers (or search heads) and the master when attempting to join the cluster. Note that the string that appears in system/local/server.conf will be the encrypted version of the original. This is done using the host's splunk.secret as a salt, so it will vary host to host. What matters is that the plain text version of the key is the same between hosts. Caution: do not leave this as the default value for a customer. Insecure or default credentials mean anyone could maliciously add themselves to the cluster and search any data they liked, regardless of the desired access permissions.

## Add the Search Head

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### Connect the Search Head to the Indexer Cluster

Next you will be connecting a search head to the cluster master allowing the search head to search the clustered indexers. When you contact a cluster master for search information, it provides an up-to-date list of the peers in case one goes down. In addition, should another indexer be added, no additional change would be required on the SH to add the new host for search. It is possible for a search head to search both standalone indexers and a cluster simultaneously, but both should not be configured for the same indexer.

1. Observe the current list of search peers (Settings -> Distributed Search -> Search Peers).

You should see no search peers currently connected to the search head.

2. Utilizing the base configurations (link above), identify the appropriate configuration app for a search head utilizing clustered search.
3. Update the app as needed to reflect the mode and the location of the master.
4. Restart Splunk on your search head.

## Check Your Work

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1. Look at the distributed search peers list in the UI after the restart. How many do you see?  
\_\_\_\_\_
2. On the Cluster Master's UI, observe that your host is now listed as a search head against the cluster.  
What is the other listed search head? \_\_\_\_\_
3. Run a search against data in the "onboarding" index over all time.  
What are the values of splunk\_server identified for the returned events? \_\_\_\_\_

If your configuration is not returning the expected results, troubleshoot by isolating the issue.

Ensure that you have the right value for the `pass4SymmKey` which should match throughout the indexer cluster and on the cluster master. This is a security mechanism to ensure that no one can connect their search head to your indexing cluster, which would allow them to bypass access restrictions on existing search heads.

### “Contact Splunk Professional Services”

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#### Migrate buckets from a non-clustered environment into the Indexer Cluster

There are several situations in the Splunk documentation ([docs.splunk.com](https://docs.splunk.com)) that provide the message "Contact Splunk Professional Services" as the "solution" to the issue at hand. One of these relates to the ability to take data, indexed in a standalone way, and cause that data to be replicated across the members of an indexing cluster. That is, customers want the ability to safeguard their existing data after setting up indexer clustering. Now is your chance to practice this.

An indexer in a cluster considers a bucket valid for replication if it has a server GUID affixed to the end of the bucket name. Without this, it is considered a single-instance bucket. A cluster with a replication factor of 1:1 will not do any replication at that time but will affix the GUID of the indexer to the bucket names, meaning that a later change in replication factor will trigger duplication "bucket fixup" for the buckets created during that time.

To force this behavior, we must add the GUID to the end of all of the buckets.

1. Connect to `idx1`.
2. Change directory into the `$SPLUNK_DB/onboarding/db` directory.

This is the homePath for the onboarding index, where hot and warm buckets are stored. This host should have been restarted a number of times before reaching this point, so there should be no hot buckets (with a directory name like "hot\_v1\_5") present. If there are, proceed to step otherwise skip to step 5.

3. Force the indexer to roll the buckets from hot to warm:  
`splunk _internal call /data/indexes/<index_name>/roll-hot-buckets`

Renaming buckets while Splunk is running is ill-advised. You're far less likely to have any issue by first stopping Splunk.

4. Stop Splunk now.
5. Identify the GUID of the single instance. It is recorded in `$SPLUNK_HOME/etc/instance.cfg`.
6. For each directory in the `homePath` of the onboarding index, rename the directory to append an underscore (`_`) followed by the GUID from step 5.
7. Start Splunk.

Why wouldn't this have worked to roll buckets from hot to warm? Why was a separate step required for that?

### Check Your Work

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1. Go to the web GUI of your cluster master instance.
2. Visit the Settings -> Indexer Clustering page, on the "Indexes" tab.

You should see your Onboarding index and some number of buckets automatically replicated.

3. Go to the web GUI of your search head instance.
4. Run the search `"| dbinspect index=onboarding"` over all time.

Note the path names for some of the buckets starts with `"rb_"` for the bucket directory's name. You should also see that the ID column is repeated.

### Troubleshooting Suggestions

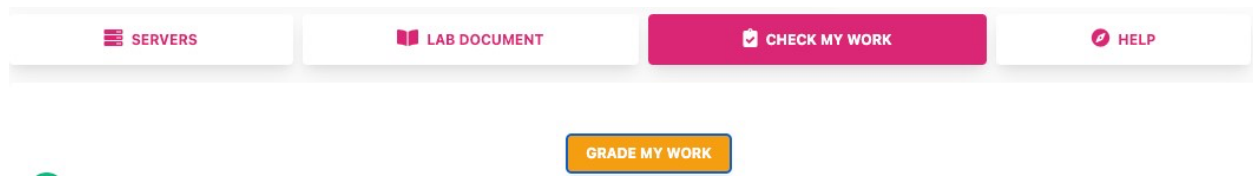
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If your configuration is not returning the expected results, troubleshoot by isolating the issue. Verify that the index definition for your onboarding index expects to be replicated. Run `bttool` to list the contents of the indexes config, focusing on the onboarding index, and validate that `repFactor` says "auto".

### Grading Your Work

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Grading of the lab is done through the same interface where your student instances are displayed.



Grading of the labs is an intensive process involving many calls to the instances. This may take several minutes to complete. Please be patient. If the grading does not complete in 5 minutes, please reach out to [ellearn@splunk.com](mailto:ellearn@splunk.com) for assistance.

For best results, make sure you complete all of the lab steps before grading the lab.

1. Click the Check My Work Tab.
2. Click the Grade My Work button.

After a few minutes, your results will be displayed.

You must pass all graded items to receive credit for the lab. Resolve any failed grading items and re-issue the grading check.

If you pass all of the lab grading checks, the Learning Management System (LMS) will terminate the browser window. Log back into the LMS to view your completed status for the lab.